

Faculty of Computers and Information Technology (FCIT)

A System for Early Detection of Gas Leaks and Fires with IoT Alerts

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DECLARATION

I hereby declare that this project report is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at University of Tabuk or other institutions.

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Specially Dedicated To

We would like to express our sincere thanks to our parents and friends for their endless encouragement and belief in us. Your constant motivation and emotional support have been the driving force that kept us moving forward, and for that, we are truly grateful.

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A System for Early Detection of Gas Leaks and Fires with IoT Alerts

Abstract

Gas leaks and fire outbreaks remain among the most dangerous hazards in both residential and industrial settings, posing serious risks to human life, property, and the environment. Traditional alarm systems, although widely available, often suffer from key limitations: many are either too expensive for widespread adoption or restricted to local alarms that depend on someone being physically present to respond. Such shortcomings can delay timely interventions and worsen the impact of an emergency.

This project presents the design and implementation of an early detection and alert system that is affordable, reliable, and user-friendly. The system employs a combination of gas, smoke, and temperature sensors to provide continuous monitoring of the surrounding environment. Once abnormal or hazardous conditions are detected, the system initiates a two-level response, level 1 consists of immediate on-site alerts using audible buzzers and visual indicators to warn occupants, and level 2 consists of real-time notifications sent to users' smartphones through IoT connectivity, enabling remote awareness and quick decision-making.

The integration of low-cost sensors with IoT technology not only ensures a practical and scalable solution but also enhances accessibility for households, small businesses, and industrial operators alike. By reducing detection-to-response time and extending alerts beyond the local premises, the proposed system contributes significantly to risk mitigation. Ultimately, it aims to minimize casualties, reduce property loss, and strengthen safety standards in everyday environments.

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IoT Internet of Things

ESP32 Espressif 32-bit Microcontroller

DHT22 Digital Humidity and Temperature Sensor

OLED Organic Light Emitting Diode

Chapter 1
INTRODUCTION

1.1 Background

Gas leaks and fires represent some of the most persistent and destructive hazards affecting both residential and industrial environments. Their occurrence is often sudden, and the consequences can range from property damage to large-scale human casualties. In domestic settings such as homes, apartments, and restaurants, small gas leaks may go undetected for hours or even days. If unnoticed, these leaks can accumulate and eventually trigger devastating fires or explosions. In industrial plants, where large volumes of flammable gases and chemicals are routinely handled, even a minor leak can escalate into catastrophic incidents involving fires, massive explosions, or the release of toxic clouds that endanger workers and surrounding communities.

The seriousness of this issue is clearly illustrated by statistical evidence. A historical analysis of industrial accidents in Saudi Arabia revealed that 54% of recorded incidents were fire-related, 25.4% involved the release of toxic clouds, and 20.6% resulted in explosions. The majority of these cases were concentrated in major industrial hubs such as Riyadh, Jeddah, and Jubail (ResearchGate, 2023). On the residential side, data reported by the *Saudi Gazette* in 2018 showed that fires caused at least 90 fatalities and 1,364 injuries within that year alone. The financial impact was also substantial, with losses in residential buildings estimated to exceed 26.5 million SAR. These numbers underscore the dual challenge: the widespread occurrence of accidents across both private and industrial sectors, and the heavy human and financial toll they impose on society.

Recognizing the urgency of this threat, national authorities have taken concrete steps to strengthen safety regulations. The Saudi Civil Defense, for instance, announced that beginning in 2024, the installation of early fire alarm systems will become mandatory in all residential buildings. The primary objective of this regulation is to reduce the average emergency response time from six minutes to just four minutes, thereby improving the chances of saving lives and minimizing damage (Argaam, 2022). Beyond compliance, these measures highlight a broader need for proactive safety systems—solutions that are not only technically reliable but also affordable and easy to deploy in a wide range of environments.

Against this backdrop, the design of advanced yet cost-effective early detection systems that integrate modern sensing and IoT technologies becomes not merely an option but a pressing necessity. Such systems can bridge the gap between existing high-cost commercial solutions and the urgent need for accessible, scalable technologies that protect lives and property on both small and large scales.

1.2 Problem overview

Although conventional fire and gas detectors are widely available on the market, many of these systems present significant limitations that reduce their overall effectiveness. Most commercial devices rely primarily on audible alarms such as buzzers or sirens. While this form of alert can be effective when occupants are present in the building, it becomes largely useless if no one is nearby to hear or respond. In residential environments, this means that hazards may go unnoticed until the fire or gas leak escalates, often when it is already too late to prevent severe damage. Similarly, in industrial facilities, relying solely on local alerts leaves critical safety decisions dependent on the immediate presence of staff, which is not always guaranteed.

Another major limitation lies in the cost of advanced detection systems. High-end solutions that integrate networked alarms and remote notifications tend to be expensive and require professional installation and maintenance. This makes them less accessible to households, small businesses, or community facilities that may lack the financial resources for such investments. Consequently, many users continue to depend on low-cost but less effective standalone detectors, which provide minimal protection against fast-moving emergencies.

The lack of integration with modern communication technologies further widens this gap. In an increasingly connected world, safety systems that fail to provide remote, real-time notifications place users at a disadvantage. Delays in detecting and responding to incidents directly increase the risk of casualties, property damage, and financial loss.

Therefore, there is a clear and urgent need for detection systems that combine affordability, high reliability, and IoT-enabled remote alerts. A solution that brings together low-cost sensors, continuous monitoring, and smartphone-based notifications can ensure timely responses regardless of the user's location. Addressing this problem will not only improve individual safety but also strengthen community resilience against fire and gas hazards.

1.3 Aim and Objectives

The overarching aim of this project is to design and present an early warning system for gas leaks and fires that enhances safety by enabling timely detection and effective alerts. The system is intended to overcome the limitations of conventional detectors by combining local alarm mechanisms with IoT-based remote notifications, ensuring that users are made aware of hazardous conditions regardless of their location.

To achieve this aim, the project sets out the following specific objectives:

1. **To integrate gas, smoke, and temperature sensors into a unified system.**

   This ensures comprehensive environmental monitoring, as each type of sensor addresses a different hazard. By combining these inputs, the system provides a more reliable and accurate detection mechanism than single-function detectors.

2. **To activate both audible and visual alarms immediately upon detecting hazards.**

   Local alerts such as buzzers and indicator lights are essential for warning nearby occupants, giving them the opportunity to take quick action before the situation escalates.

3. **To provide real-time notifications to users' smartphones through IoT technology.**

   Remote alerts enable users to respond promptly even if they are away from the site, bridging the gap left by conventional systems that depend solely on local alarms. This feature is critical for minimizing delays in emergency response.

4. **To ensure that the proposed solution remains affordable, reliable, and easy to install.**

   By focusing on cost-effectiveness and simplicity, the system becomes accessible not only to industrial operators but also to households and small businesses, encouraging broader adoption and strengthening overall community safety.

In essence, these objectives collectively aim to deliver a safety solution that is practical, scalable, and capable of reducing both human and material losses in the face of fire and gas-related hazards.

1.4 Scope of the Project

The scope of this project is intentionally defined to focus on early detection and alerting, rather than on active fire suppression or containment. The proposed system will be designed for small-scale environments such as homes, apartments, restaurants, and small workshops, where affordability, ease of installation, and reliability are essential.

The project will concentrate on the following aspects:

- **Hazard Detection:** Using integrated gas, smoke, and temperature sensors to continuously monitor the environment for signs of leaks or fire-related risks.
- **Local Alerts:** Triggering audible and visual alarms within the premises to immediately warn occupants and allow for quick evacuation or corrective action.
- **Remote Notifications:** Employing IoT-based communication to send real-time alerts to users' smartphones, ensuring awareness even when occupants are not physically present at the site.

It is important to note that the system does not include active suppression mechanisms, such as sprinklers, automatic extinguishers, or advanced industrial-scale safety controls. Instead, it prioritizes rapid detection and communication, recognizing that timely awareness is often the most critical factor in reducing harm and preventing escalation.

By clearly defining these boundaries, the project remains feasible, cost-effective, and highly relevant to the needs of everyday users while laying a foundation for possible future enhancements, such as integration with suppression systems or smart home automation platforms.

1.5 Report Organization

This report is structured into five chapters to provide a systematic presentation of the project, from background research to conclusions and recommendations. Each chapter serves a specific purpose in guiding the reader through the development and evaluation of the proposed early warning system:

- **Chapter 1 – Introduction:** Presents the background of the study, identifies the problem, and defines the project's aim, specific objectives, and scope. This chapter sets the context and rationale for the research.
- **Chapter 2 – Related Work:** Reviews existing literature, studies, and commercial detection systems for gas leaks and fires. It highlights the limitations of current solutions and identifies gaps that the proposed system aims to address.
- **Chapter 3 – Methodology:** Describes the design and architecture of the proposed system, detailing its hardware components, sensors, IoT integration, and operational principles. This chapter explains how the system functions to detect hazards and issue alerts.
- **Chapter 4 – Results and Discussion:** Presents the outcomes of system implementation and testing. The chapter analyzes the performance, reliability, and effectiveness of the proposed solution, comparing it to existing alternatives where applicable.
- **Chapter 5 – Conclusion and Future Work:** Summarizes the key findings, highlights the contributions of the project, and discusses potential improvements and future directions for research and development in early hazard detection systems.

Chapter 2

RELATED WORK

2.1 Introduction

Fire detection and gas leakage monitoring systems have received growing attention over the past decades due to their crucial role in ensuring safety. From the early standalone smoke alarms introduced in the 1970s to today's sophisticated IoT-enabled systems, the evolution of detection technology reflects the increasing need for reliable, accessible, and affordable solutions. While considerable progress has been made, challenges remain in creating systems that balance cost, effectiveness, and usability, particularly for small-scale applications such as homes, small restaurants, and offices.

This chapter reviews relevant academic literature, commercial systems, and experimental approaches. It provides an in-depth discussion of their strengths and limitations, followed by an analysis of the gaps that the proposed project aims to address.

2.2 Traditional Fire and Gas Detection Systems

Conventional detection systems typically rely on standalone sensors, including smoke detectors, heat sensors, and chemical gas alarms.

2.2.1 Gas Detectors

- MQ-series sensors (MQ-2, MQ-6, MQ-135) are popular for detecting gases like methane, butane, carbon monoxide, and LPG. They are inexpensive and widely used in research and educational projects.
- Limitations include sensitivity to temperature, humidity, and the need for frequent recalibration.

2.2.2 Heat Detectors

- Function by monitoring abnormal rises in temperature.
- Reliable in harsh environments where smoke detectors may fail (dusty or humid areas).
- However, they are slower to respond compared to smoke alarms.

Strengths of traditional systems:

- Affordable and widely available.

- Simple to install and maintain.
- Effective for localized environments.

Limitations:

- Only provide local alerts without remote communication.
- Depend on human presence to respond.
- Reduced effectiveness in large or unmonitored areas.

2.3 IoT-Based Detection Systems

Recent advancements in IoT have enabled systems that combine sensors with wireless communication, cloud servers, and mobile applications.

Case Studies:

- IoT Fire Safety (UTHM, 2020): Used ESP32 with MQTT protocol for real-time monitoring. Users were alerted through mobile applications. However, the system heavily relied on stable Wi-Fi connectivity.
- LPG Leakage Detector (IJRASET, 2021): Employed MQ-6 with Arduino and GSM modules to send SMS alerts and activate fans. While practical, it had limited coverage and relatively higher setup costs.
- Cloud-Enabled Smoke Detection (IEEE, 2019): Integrated smoke sensors with cloud servers for continuous monitoring and historical data storage. The system allowed remote management but introduced privacy and subscription cost concerns.

Strengths:

- Remote notifications via SMS, apps, or calls.
- Real-time monitoring across multiple sites.
- Scalable for both residential and industrial use.

Limitations:

- Dependence on stable internet or GSM signals.
- Higher initial and operational costs.
- Technical complexity in setup and maintenance.

2.4 Hybrid Systems and Advanced Approaches

Hybrid systems combine the affordability of traditional devices with the extended features of IoT, while advanced research explores intelligent detection models.

Examples:

- Autonomous Fire Safety (MEC, 2022): Combined MQ sensors with Raspberry Pi. It not only sent alerts but also controlled exhaust fans and shutoff valves, making it more preventive.
- AI-Enhanced Detection (Springer, 2021): Applied machine learning to reduce false alarms by distinguishing between hazardous smoke and harmless cooking emissions. While highly accurate, these systems remain experimental and resource-intensive.

Advantages:

- Balance between cost and functionality.
- Ability to perform preventive actions in addition to sending alerts.

Drawbacks:

- Require more complex designs and software.
- Limited commercial adoption due to cost and technical barriers.

2.5 Commercial Systems

Commercial fire and gas detection solutions vary from low-cost household devices to industrial-grade platforms.

2.5.1 Household Detectors

- Kidde and First Alert provide standalone smoke and gas detectors. These are affordable and battery-operated but limited to local alerts.

2.5.2 Smart Home Detectors

- Google Nest Protect offers Wi-Fi connectivity, mobile notifications, self-testing, and voice alerts.
- X-Sense Smart Smoke & CO Alarm supports smartphone apps and integrates with smart home platforms.
- These devices are effective but costlier (typically $100–$150 per unit).

2.5.3 Industrial-Grade Systems

- Companies like Honeywell and Siemens provide large-scale solutions with high sensitivity, redundancy, and automatic suppression (sprinklers, ventilation systems).
- Designed for factories, oil plants, and high-risk environments.
- While robust, they are prohibitively expensive for small businesses and households.

Tables 2.8.1 and 2.8.2 present a comparative overview of existing fire and gas detection systems, highlighting both research-based approaches and commercially available solutions. The analysis of these tables provides insight into the evolution of detection technologies, their respective advantages, and the challenges that motivate the development of the proposed system.

From Table 2.8.1, it is evident that traditional systems remain the most cost-effective and straightforward option. Their primary advantage lies in their simplicity and affordability, making them suitable for basic household or small-space applications. However, their functionality is limited to local alerts and manual intervention, with minimal scalability and no remote monitoring capabilities (IEEE, 2018; Elsevier, 2019).

In contrast, IoT-based systems extend functionality by enabling remote monitoring, real-time alerts, and system scalability through internet connectivity. These systems significantly enhance safety management, but they introduce dependency on stable internet connections and often come with higher implementation costs (UTHM, 2020; IJRASET, 2021).

Hybrid systems represent an intermediate stage, integrating both local and remote alert mechanisms, often coupled with automated control actions such as shutting off gas valves or activating sprinklers. While this configuration offers improved safety and flexibility, it also increases system complexity and cost (MEC, 2022).

Finally, AI-based systems mark the most advanced generation of detection technologies. By leveraging machine learning and computer vision, these systems achieve superior accuracy and significantly reduce false alarms. However, most remain in experimental stages and demand high computational resources, limiting their current practicality for low-cost or small-scale environments (Springer, 2021).

The commercial comparison in Table 2.8.2 reinforces similar trends. Household systems such as Kidde and First Alert are affordable and user-friendly but are restricted to local alarm functionality, lacking connectivity or automation features. Smart home solutions, including Nest Protect and X-Sense, provide enhanced intelligence through remote alerts, self-testing, and integration with home automation ecosystems. However, these advantages come at the expense of higher cost and reliance on internet

connectivity. Meanwhile, industrial-grade systems from Honeywell and Siemens demonstrate exceptional reliability, precision, and certification compliance, making them ideal for large-scale or critical environments, but their complexity and cost render them impractical for everyday household use.

Overall, the analysis underscores a persistent trade-off between functionality, cost, and accessibility across existing systems. Traditional and household systems are affordable but limited in intelligence, while IoT, hybrid, and AI-based solutions provide greater functionality at the cost of affordability and simplicity. This technological gap highlights the need for a balanced, cost-effective, and intelligent fire and gas detection system, capable of delivering real-time alerts, local actuation, and reliable performance without excessive financial or infrastructure demands.

Table 2.8.1 Comparative Summary

Type	Strengths	Limitations	References
Traditional	Low cost, simple	Local only, less scalable	IEEE (2018), Elsevier (2019)
IoT-based	Remote alerts, scalable	Internet-dependent, costly	UTHM (2020), IJRASET (2021)
Hybrid	Local + remote + actions	Complex, higher cost	MEC (2022)
AI-based	High accuracy, fewer false alarms	Experimental, resource-heavy	Springer (2021)

Table 2.8.2 Commercial Systems

Category	Examples	Strengths	Limitations
Household	Kidde, First Alert	Affordable, simple	Only local alerts
Smart Home	Nest Protect, X-Sense	Remote alerts, self-testing	Expensive, internet-dependent
Industrial	Honeywell, Siemens	Reliable, certified	High cost, complex setup

2.6 Gap Analysis

The review highlights several key gaps:

- Integration Gap: Lack of affordable systems combining smoke, gas, and heat detection into one compact solution.
- Accessibility Gap: Limited options tailored for homes, small businesses, and restaurants.
- Usability Gap: Many systems are either too simple with limited features or too advanced with high complexity.

The proposed project addresses these gaps by offering:

- A low-cost, multi-sensor system (gas, smoke, and temperature).
- Dual alerts: local alarms and smartphone notifications.
- User-friendly operations suitable for small-scale environments.

2.7 Conclusion

In conclusion, while fire and gas detection systems have evolved significantly, current solutions are either too limited in functionality or too expensive for everyday users. Research studies and commercial systems demonstrate potential but leave gaps in affordability and usability. The proposed project seeks to provide a balanced solution that is cost-effective, reliable, and practical for small environments, ultimately improving safety without imposing financial or technical burdens.

In the following chapter, the methodology adopted for designing and implementing the proposed system is presented in detail.

Chapter 3
DESIGN

3.1 Introduction

This chapter describes the overall design, structure, and operational methodology of the proposed Fire and Gas Detection and Alert System. It focuses on the hardware and software integration, system architecture, circuit design, and functional flow of the system. The project aims to detect hazardous conditions such as gas leaks, smoke, and excessive heat, then issue immediate local and remote alerts to ensure safety and minimize potential damage.

The system integrates multiple sensors with a microcontroller and IoT platform, enabling both on-site alarm activation and remote notifications through mobile devices. This design ensures real-time monitoring, reliability, and cost-effectiveness for domestic and small industrial environments.

3.2 Circuit Diagram

The circuit diagram for the Gas Detection and Monitoring System, shown in Figure 3.1 (Circuit Diagram), illustrates the electrical connections between the ESP32 microcontroller, sensors, and output components.

The MQ-2 gas sensor is connected to an analog input pin of the ESP32 to detect gas concentration levels in the environment, while the DHT22 sensor is connected to a digital pin to measure ambient temperature and humidity. A buzzer is connected to a digital output pin to provide an audible alert during warning and danger conditions. The 0.96-inch OLED display is connected using I2C communication to display real-time sensor readings and system status messages. LED indicators are also connected to visually represent the system states (Normal, Warning, and Danger).

To improve circuit stability and protect the microcontroller, an NPN transistor is used to control the buzzer. The transistor acts as an electronic switch that allows the buzzer to operate without placing excessive load on the ESP32 GPIO pins. In addition, a 100µF capacitor is added to stabilize the power supply and reduce voltage drops that may occur when components draw sudden current. A 100nF capacitor is also included to filter high-frequency electrical noise and improve overall signal stability in the circuit. Furthermore, the system supports IoT functionality using the built-in Wi-Fi module of the ESP32. Sensor data and alert notifications can be sent to a mobile application or cloud platform, allowing remote monitoring and instant notifications

when abnormal gas or temperature levels are detected. The ESP32 acts as the main controller that reads environmental data, processes the inputs, and activates local alerts as well as IoT notifications, ensuring a smart, reliable, and continuous safety monitoring system.

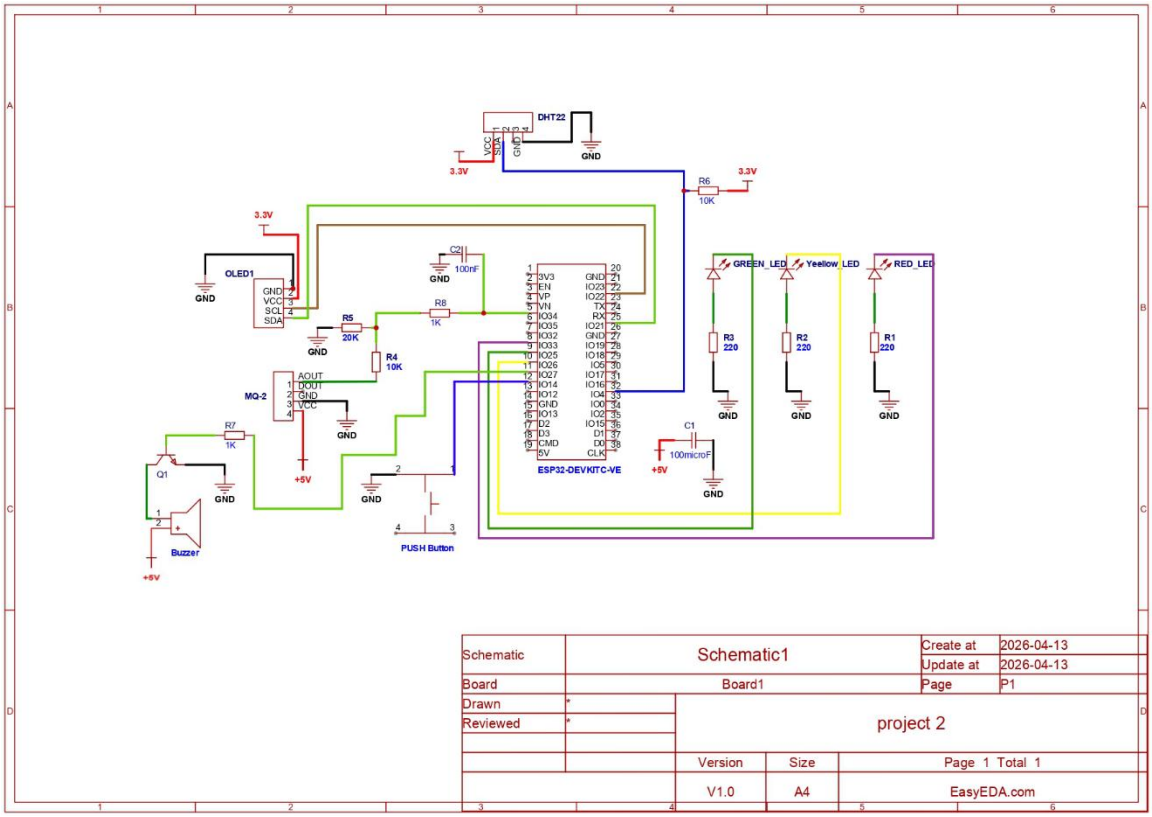


Figure 3.1: Circuit Diagram

Table 3.1 presents the hardware connections used in the circuit shown in Figure 3.1 (Circuit Diagram). It summarizes the ESP32 GPIO assignments, additional electronic components such as resistors and capacitors, and the function of each connection within the system.

Table 3.1: Hardware Connections Used in Figure 3.1 (Circuit Diagram)

Component	ESP32 Pin	Additional Components	Description
MQ-2 Gas Sensor (AOUT)	GPIO34	Voltage divider (10kΩ, 20kΩ), 1kΩ resistor, 100nF capacitor	Reads analog gas concentration
MQ-2 Power	5V / GND	100µF capacitor	Provides stable power supply
DHT22 Sensor (DATA)	GPIO4	10kΩ pull-up resistor	Measures temperature and humidity
OLED Display (SDA)	GPIO21	I2C interface	Data line for OLED communication
OLED Display (SCL)	GPIO22	I2C interface	Clock line for OLED communication
Green LED	GPIO25	220Ω resistor	Indicates NORMAL state
Yellow LED	GPIO26	220Ω resistor	Indicates WARNING state
Red LED	GPIO33	220Ω resistor	Indicates DANGER state
Buzzer	GPIO27	NPN transistor + 1kΩ resistor	Generates audible alarm
Push Button	GPIO14	Internal pull-up	Used to mute the alarm
Power Supply	3.3V / 5V	Common GND	Powers the system

3.3 System Program Code and Implementation

```
#define BLYNK_TEMPLATE_ID "TMPL6DYSAszvR"
#define BLYNK_TEMPLATE_NAME "Quickstart Template"
#define BLYNK_PRINT Serial

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

char auth[] = "000000000000000";
char ssid[] = "00";
char pass[] = "11223344";

#define DHTPIN 4
#define DHTTYPE DHT22
#define PIN_GAS    34
#define LED_G      25
#define LED_Y      26
#define LED_R      33
#define BUZZER     27
#define MUTE_BTN   14

DHT dht(DHTPIN, DHTTYPE);
Adafruit_SSD1306 display(128, 64, &Wire, -1);

// ----------- Variables -----------
bool alertSent = false;
bool gasBaseReady = false;
int gasBase = 0;

bool muted = false;
bool lastBtnState = HIGH;

unsigned long bootMs = 0;
unsigned long lastSend = 0;

// ----------- Thresholds -----------
const int GAS_WARN_DELTA   = 167;
const int GAS_DANGER_DELTA = 253;
const float TEMP_WARN_C    = 50.0;
const float TEMP_DANGER_C  = 60.0;

// ----------- System State -----------
enum State { ST_NORMAL, ST_WARN, ST_DANGER };
State state = ST_NORMAL;

// ----------- Blynk Button (Mute Control) -----------
```

```
BLYNK_WRITE(V1) {
  muted = param.asInt(); // 1 = Mute ON
}

// ----------- Functions -----------
void setLeds(State s) {
  digitalWrite(LED_G, s == ST_NORMAL);
  digitalWrite(LED_Y, s == ST_WARN);
  digitalWrite(LED_R, s == ST_DANGER);
}

void alarmOn() {
  if (!muted) tone(BUZZER, 2000); // Activate buzzer if not muted
  else noTone(BUZZER);            // Stop buzzer if muted
}

void alarmOff() {
  noTone(BUZZER); // Ensure buzzer is OFF
}

void setup() {
  Serial.begin(115200);
  bootMs = millis();

  pinMode(LED_G, OUTPUT);
  pinMode(LED_Y, OUTPUT);
  pinMode(LED_R, OUTPUT);
  pinMode(BUZZER, OUTPUT);
  pinMode(MUTE_BTN, INPUT_PULLUP);

  setLeds(ST_NORMAL);
  alarmOff();

  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);

  dht.begin();

  // Connect to WiFi without blocking forever
  WiFi.begin(ssid, pass);
  int timeout = 0;
  while (WiFi.status() != WL_CONNECTED && timeout < 20) {
    delay(500);
    Serial.print(".");
    timeout++;
  }

  if (WiFi.status() == WL_CONNECTED) {
    Serial.println("WiFi Connected");
```

```
    Blynk.config(auth);
    Blynk.connect();
  } else {
    Serial.println("WiFi Failed");
  }
}

void loop() {

  // Run Blynk only if WiFi is connected
  if (WiFi.status() == WL_CONNECTED) {
    Blynk.run();
  }

  // Read sensors
  float t = dht.readTemperature();
  float h = dht.readHumidity();
  int gasRaw = analogRead(PIN_GAS);

  // MQ-2 warm-up period (ignore readings during this time)
  if (millis() - bootMs < 20000) {
    display.clearDisplay();
    display.setCursor(0, 0);
    display.println("MQ2 WARMING UP...");
    display.display();
    return;
  }

  // Set baseline (clean air reference)
  if (!gasBaseReady) {
    gasBase = gasRaw;
    gasBaseReady = true;
  }

  // Calculate gas difference from baseline
  int gasDelta = gasRaw - gasBase;
  if (gasDelta < 0) gasDelta = 0;

  // Determine system state
  if (gasDelta >= GAS_DANGER_DELTA || t >= TEMP_DANGER_C) state =
ST_DANGER;
  else if (gasDelta >= GAS_WARN_DELTA || t >= TEMP_WARN_C) state =
ST_WARN;
  else state = ST_NORMAL;

  // Send notification once when entering danger state
  if (state == ST_DANGER && !alertSent) {
    Blynk.logEvent("gas_alert");
    alertSent = true;
  }
```

```
if (state != ST_DANGER) alertSent = false;

// Physical mute button handling (edge detection)
bool currentBtn = digitalRead(MUTE_BTN);
if (lastBtnState == HIGH && currentBtn == LOW && state == ST_DANGER) {
  muted = !muted; // Toggle mute
}
lastBtnState = currentBtn;

// Control outputs
setLeds(state);
if (state == ST_DANGER) alarmOn();
else alarmOff();

// OLED display
display.clearDisplay();
display.setCursor(0, 0);

display.print("Temp: ");
if(isnan(t)) display.println("Err"); else {display.print(t, 1); display.println(" C");}

display.print("Hum:  ");
if(isnan(h)) display.println("Err"); else {display.print(h, 1); display.println(" %");}

display.print("Gas:  "); display.println(gasDelta);

display.print("Stat: ");
if(state == ST_NORMAL) display.println("NORMAL");
else if(state == ST_WARN) display.println("WARNING");
else display.println("DANGER!!");

if (muted) display.println("MUTED");

display.display();

// Send data to Blynk every 1 second
if (millis() - lastSend > 1000 && WiFi.status() == WL_CONNECTED) {

  Blynk.virtualWrite(V2, t);
  Blynk.virtualWrite(V3, h);
  Blynk.virtualWrite(V4, gasDelta);

  if(state == ST_NORMAL) Blynk.virtualWrite(V5, "NORMAL");
  else if(state == ST_WARN) Blynk.virtualWrite(V5, "WARNING");
  else Blynk.virtualWrite(V5, "DANGER");

  lastSend = millis();
}

delay(200);
```

}

3.3.1 Detailed Program Code Description

This program implements a smart gas and temperature monitoring system using an ESP32 microcontroller, an MQ-2 gas sensor, a DHT22 temperature and humidity sensor, an OLED display, LEDs, a buzzer, and a mute button. In addition, the system integrates IoT functionality using the Blynk IoT platform to enable remote monitoring and real-time notifications.
The system detects abnormal environmental conditions and responds accordingly using a Finite State Machine (FSM).

- **Libraries and Initialization**

The code includes the following libraries:

- WiFi.h → Handles WiFi connection
- BlynkSimpleEsp32.h → Enables communication with the Blynk platform
- Wire.h → I2C communication
- Adafruit_GFX.h and Adafruit_SSD1306.h → OLED display control
- DHT.h → Temperature and humidity sensor handling

The OLED display is initialized using I2C with address 0x3C.
The DHT22 sensor is connected to GPIO 4.
The MQ-2 gas sensor analog output is connected to GPIO 34 (ADC1).

- **System Components and Pin Configuration**

The system includes the following components:

- Green LED → Indicates NORMAL state
- Yellow LED → Indicates WARNING state
- Red LED → Indicates DANGER state
- Buzzer → Provides audible alarm
- Push Button (GPIO 14) → Used to mute the alarm
- WiFi Module (ESP32 built-in) → Enables IoT communication

All pins are configured in the setup() function using pinMode().

- **WiFi and Blynk Initialization**

The system attempts to connect to a WiFi network during startup. A timeout mechanism is used to prevent the system from getting stuck if the connection fails.
If WiFi is successfully connected:

- The system initializes Blynk using Blynk.config()
- Establishes a connection using Blynk.connect()

If WiFi fails:

- The system continues operating locally without remote features

- **Sensor Data Acquisition**

The system continuously reads:

- Temperature and humidity from the DHT22 sensor
- Gas level from the MQ-2 sensor using analog input

A warm-up period of 20 seconds is implemented for the MQ-2 sensor. During this phase:

- Gas readings are ignored
- The OLED displays a “Warming Up” message

- The system remains in NORMAL state

After warm-up:

- The first gas reading is stored as a baseline value
- Future readings are compared against this baseline

Gas level is calculated as:

Gas Delta = Current Reading − Baseline

- **Thresholds and Decision Logic**

The system defines two threshold levels:

- Warning Threshold
- Danger Threshold

For both:

- Gas level (delta value)
- Temperature

State determination rules:

- If gas or temperature ≥ danger threshold → DANGER
- Else if gas or temperature ≥ warning threshold → WARNING
- Otherwise → NORMAL

The DANGER state has the highest priority.

- **Finite State Machine (FSM)**

The system operates using three main states:

- ST_NORMAL
- ST_WARN
- ST_DANGER

The FSM continuously evaluates sensor readings and transitions between states based on threshold conditions.

- **Alarm and LED Control**

The system reacts based on the current state:

- **NORMAL**
  - Green LED ON
  - Buzzer OFF
- **WARNING**
  - Yellow LED ON
  - Buzzer OFF
- **DANGER**
  - Red LED ON
  - Buzzer ON

The buzzer operates using a tone signal and is activated only during the danger state.

- **Mute Functionality**

The system provides two methods to mute the buzzer:

- 1. Physical Button
- When pressed during the DANGER state
- Toggles the mute status (ON/OFF)
- 2. Blynk Button (Virtual Pin V1)
- Allows remote mute control from the mobile app

When muted:

- The buzzer stops immediately
- The system continues monitoring

- **Notification System (Blynk IoT)**

When the system enters the DANGER **state**:

- A notification is sent using:

Blynk.logEvent("gas_alert");

To avoid repeated alerts:

- A flag (alertSent) ensures the notification is sent only once per danger event

- **OLED Display Output**

The OLED display shows real-time system data:

- Temperature (°C)
- Humidity (%)
- Gas level (delta value)
- System state (NORMAL / WARNING / DANGER)

If mute is activated:

- The display shows **"**MUTED**"**

- **Remote Monitoring via Blynk**

The system sends data to Blynk every 1 second:

- V2 → Temperature
- V3 → Humidity
- V4 → Gas Level
- V5 → System State

This allows real-time monitoring through the mobile application dashboard.

- **Overall System Behavior**

The system operates continuously in a loop:

1. Reads sensor values
2. Applies warm-up and baseline logic
3. Calculates gas deviation
4. Determines system state
5. Controls LEDs and buzzer
6. Displays data on OLED
7. Sends data to Blynk
8. Sends alerts if danger is detected

- **System Advantages**
- Real-time monitoring
- Reduced false alarms using baseline calibration
- Remote notification via IoT
- User control through mute functionality
- Clear visualization through OLED and mobile app

3.4 State Machine Diagram

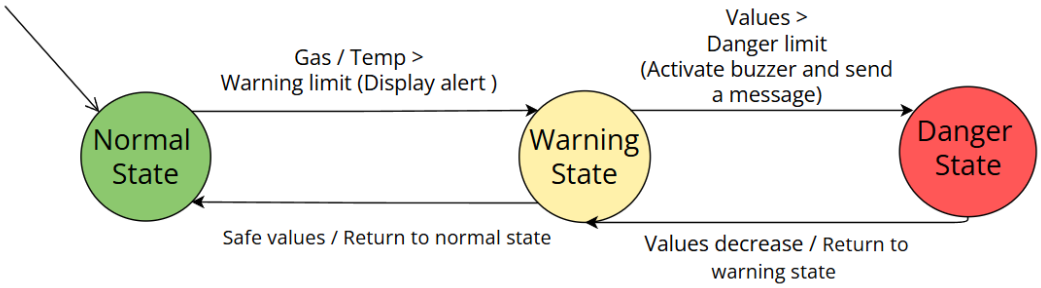


Figure 3.2: State Machine Diagram

3.5 Transitions

The system transitions between states according to predefined conditions derived from sensor data. The transition logic is summarized below in Table 3.2 (Transition).

Table 3.2 Transition

Current State	Condition / Trigger	Next State
Normal	Gas or temperature exceeds warning limit	Warning
Warning	Values exceed danger limit	Danger
Warning	Values return to normal range	Normal
Danger	Values decrease / Return to warning state	Warning

This transition structure ensures that the system responds promptly to unsafe conditions while automatically reverting to monitoring mode once the hazard has passed.

3.6 Block Diagram

The **Block Diagram** (Figure 3.3) illustrates the main functional modules of the system and their interconnections. It includes the following components:

1. **Sensor Unit:**
    - MQ-2 Gas and Smoke Sensor
    - DHT22 Temperature and Humidity Sensor
2. **Processing Unit:**
    - ESP32 Microcontroller (handles data reading, decision-making, and IoT communication).
3. **Output Unit:**
    - Buzzer for audible alerts.
    - OLED display for real-time feedback and status messages.
4. **IoT Communication Layer:**
    - Uses the built-in Wi-Fi module of ESP32 to send data and alerts to cloud services such as **Blynk**.

This modular architecture enhances reliability, allows easy scalability, and enables real-time monitoring both locally and remotely.

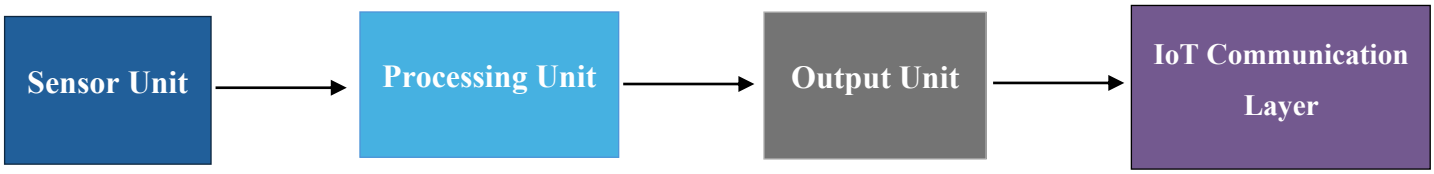


Figure 3.3 The Block Diagram

3.7 Hardware Components

3.7.1 ESP32's Wi-Fi

The ESP32 is the core processing unit and serves as the “brain” of the system. As shown in Figure 3.4 (ESP32’s Wi-Fi), it features a dual-core processor with built-in Wi-Fi and Bluetooth capabilities, making it ideal for Internet of Things (IoT) applications.

Functions in the project:

- Reads sensor data from the MQ-2 gas sensor and the DHT22 temperature sensor.
- Compares the readings with predefined safety thresholds.
- Activates the buzzer during danger conditions.
- Displays real-time readings and system messages on the OLED screen.
- Sends alert notifications to users through IoT platforms such as Blynk via Wi-Fi.

Reason for selection:

The ESP32 was chosen for its powerful processing capability, low cost, and integrated wireless communication, which simplifies system design and reduces the need for external modules.

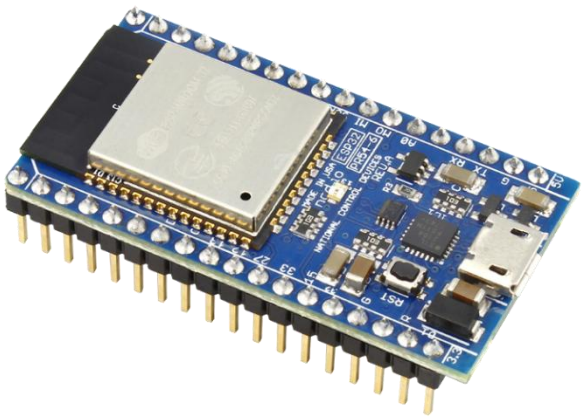


Figure 3.4 ESP32's Wi-Fi

3.7.2 MQ-2 Gas and Smoke Sensor

The MQ-2 sensor is used for detecting combustible gases such as LPG and methane, as well as smoke generated by combustion. This dual-detection capability makes it ideal for both fire detection and gas leakage monitoring applications. As shown in Figure 3.5 (MQ-2 Gas and Smoke Sensor), it consists of a sensing element made of tin dioxide (SnO_2), whose resistance varies depending on the gas or smoke concentration in the surrounding air.

Functions in the project:

- Continuously measures gas and smoke concentrations.
- Sends analog signals to the ESP32 for processing.
- Enables early detection of hazardous gas leaks or smoke to prevent fires or explosions.

Operating principle:
When the concentration of gas or smoke increases, the resistance of the sensor decreases, causing the output voltage to rise. The ESP32 reads this change and determines whether the environment is in a safe, warning, or danger state.

Reason for selection:
The MQ-2 offers high sensitivity to multiple gases and smoke, ease of integration with microcontrollers, and low cost, making it ideal for real-time fire and gas detection systems.



Figure 3.5 MQ-2 Gas and Smoke Sensor

3.7.3 DHT22 Temperature and Humidity Sensor

The DHT22 sensor measures ambient temperature and relative humidity, providing digital data directly to the ESP32 through a single-wire interface. As shown in Figure 3.6 (DHT22 Temperature and Humidity Sensor), it provides reliable environmental measurements that support both safety detection and system monitoring.

Functions in the project:

- Monitors environmental temperature to detect abnormal heat that may indicate a fire risk.
- Supports analysis of gas readings, as higher temperatures can influence gas dispersion.
- Displays real-time temperature and humidity readings on the OLED screen.

Operating principle:
The DHT22 uses a thermistor and a capacitive humidity sensor to detect environmental conditions and transmits the data as a digital signal to the ESP32 for analysis.

Reason for selection:
The DHT22 was chosen for its low cost, stability, and sufficient accuracy for small-scale environmental monitoring applications.

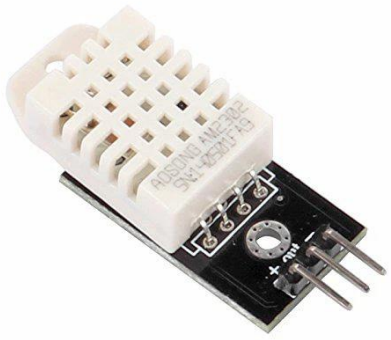


Figure 3.6 DHT22 Temperature and Humidity Sensor

3.7.4 0.96” OLED Display

The 0.96-inch OLED display is used to provide real-time visual feedback to the user. It communicates with the ESP32 using the I2C protocol, which requires only two wires (SCL and SDA). As shown in Figure 3.7 (0.96” OLED Display)**,** the display provides a compact and efficient user interface for monitoring system status and sensor readings.

Functions in the project:

- Displays live readings for gas, smoke, temperature, and humidity.
- Shows the system’s current state: Normal, Warning, or Danger.
- Displays alert messages such as “Gas Leak Detected” or “High Temperature Alert”.

Reason for selection:
The OLED display offers clear visibility, low power consumption, and a compact design, making it suitable for embedded IoT safety systems.



Figure 3.7 0.96” OLED Display

3.7.5 Buzzer

The buzzer is the system's main audible alert device. It provides an immediate, loud sound signal whenever a dangerous condition is detected. As shown in Figure 3.8 (Buzzer)**,** it serves as a critical local warning mechanism to alert nearby individuals.

Functions in the project:

- Activated automatically when gas, smoke, or temperature readings exceed danger thresholds.
- Produces an intermittent tone to alert nearby individuals of potential hazards.
- Stops automatically once conditions return to normal or when manually reset.

Reason for selection:

The buzzer is simple, inexpensive, and highly effective for drawing attention quickly during emergencies.



Figure 3.8 Buzzer

3.7.6 NPN Transistor Integration

The NPN transistor is integrated into the circuit to safely drive the buzzer using an external current path while keeping the ESP32 GPIO pin protected. As shown in Figure 3.9 (NPN Transistor Integration), the transistor operates as an electronic switch: the ESP32 provides a small control signal to the transistor base, and the transistor allows a larger current to flow through the buzzer when an alarm condition occurs.

Functions in the project:

- Acts as a switching driver to control the buzzer without overloading the ESP32 output pin.
- Provides current amplification so the buzzer can operate reliably with sufficient current.
- Protects the ESP32 GPIO by isolating the microcontroller from high-load conditions and sudden current draw.

Reason for selection:

The NPN transistor was selected because it is a low-cost and efficient solution for driving loads that require higher current than a microcontroller pin can safely supply. It improves circuit reliability and protects the ESP32 from potential damage during alarm activation.

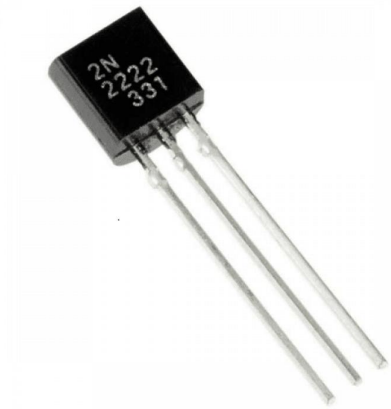


Figure 3.9 NPN Transistor Integration

3.7.7 100µF Capacitor (Electrolytic Capacitor)

The 100µF electrolytic capacitor is added to improve power stability in the system by reducing sudden voltage drops caused by rapid current changes. As shown in Figure 3.10 (100µF Capacitor (Electrolytic Capacitor)), it acts as a small energy storage element that supplies extra current when the buzzer or other components draw a sudden load, helping the ESP32 operate reliably.

Functions in the project:

- Stabilizes the power supply by smoothing voltage fluctuations on the 5V/3.3V rails.
- Reduces sudden voltage drops during buzzer activation or load changes.
- Helps prevent ESP32 resets and improves overall system reliability during operation.

Reason for selection:

The 100µF electrolytic capacitor was selected because it provides high capacitance in a compact and low-cost component. It is effective for bulk decoupling and maintaining stable voltage in embedded systems where current demand can change quickly



Figure 3.10 100µF Capacitor (Electrolytic Capacitor)

3.7.8 100nF Capacitor (Ceramic Capacitor)

The 100nF ceramic capacitor is added to the circuit for high-frequency noise filtering and signal stabilization. As shown in Figure 3.11 (100nF Capacitor (Ceramic Capacitor)), it is typically placed close to the microcontroller or sensitive signal lines to suppress fast electrical noise and reduce fluctuations that can affect sensor readings and system performance.

Functions in the project:

- Filters high-frequency electrical noise on power and signal lines.
- Reduces voltage spikes and improves signal stability for reliable sensor readings.
- Helps prevent unwanted fluctuations that may cause unstable behavior or false triggering.

Reason for selection:

The 100nF ceramic capacitor was selected because it is a standard and effective decoupling component used in embedded circuits. It is low cost, compact, and highly efficient at removing high-frequency noise, improving overall system stability.



Figure 3.11 100nF Capacitor (Ceramic Capacitor)

3.7.9 Overall Impact on System Performance

The addition of the transistor and capacitors significantly improves the system's reliability and electrical robustness. These components protect the ESP32 from overcurrent conditions, stabilize the power supply, reduce noise, and prevent unexpected resets. As a result, the overall system becomes more stable, safer, and suitable for real-world deployment.

3.7.10 Push Button

The push button is included as a user control input to manage the alarm behavior during critical conditions. As shown in Figure 3.12 (Push Button)**,** it is connected to the ESP32 as a digital input (commonly using an internal pull-up resistor), allowing the user to interact with the system when an alarm is active.

Functions in the project:

- Allows the user to mute the buzzer during the DANGER state to reduce continuous noise while responding to the emergency.
- Provides manual user interaction for alarm acknowledgment without turning off the entire system.
- Supports safer operation by enabling controlled alarm silencing while monitoring continues.

Reason for selection:

The push button was selected because it is simple, low-cost, and reliable. It provides an easy and direct way for the user to control the alarm output, improving usability and making the system more practical for real-world use.



Figure 3.12 Push Button

3.7.11 Resistors (220Ω, 1kΩ, 10kΩ, and 20kΩ)

Resistors are used in the circuit to control current flow, protect the ESP32 pins, and ensure stable and reliable sensor readings. As shown in Figure 3.13 (Resistors), different resistor values are placed in specific parts of the circuit based on each component's electrical requirements.

Functions in the project:

- 220Ω resistors (LED current limiting): Connected in series with the LEDs to limit current and protect both the LEDs and ESP32 GPIO pins from excessive current.
- 1kΩ resistor (current limiting and protection): Used to limit current and protect the ESP32. It is commonly placed at the transistor base to safely drive the buzzer, and/or used as a series resistor on the MQ-2 analog sensing path to protect the ADC input and reduce noise depending on the circuit design.
- 10kΩ resistor (pull-up / part of voltage divider): Used as a pull-up resistor to ensure stable digital signals (such as the DHT22 data line), and/or used as part of the MQ-2 voltage divider to scale the analog output safely for the ESP32 ADC.
- 20kΩ resistor (MQ-2 voltage divider): Works with the 10kΩ resistor to form a voltage divider that reduces and stabilizes the MQ-2 analog output, ensuring the ADC input remains within a safe measurement range.

Reason for selection:

These resistor values were selected because they are standard, widely available, and suitable for embedded system design. The 220Ω resistors provide safe LED operation, while the 10kΩ and 20kΩ values form an effective voltage divider and/or stable pull-up configuration for reliable readings. The 1kΩ resistor limits control/signal current and helps protect the ESP32, improving overall circuit safety and stability.

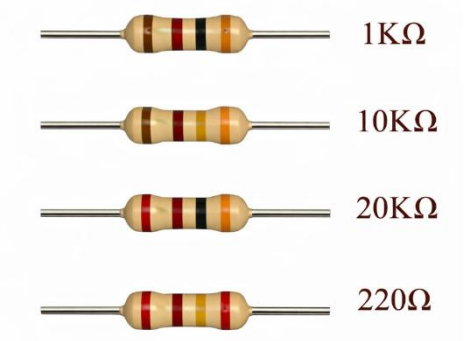


Figure 3.13 Resistors (220Ω, 1kΩ, 10kΩ, and 20kΩ)

3.8 Flowchart Diagram

The Flowchart Diagram (Figure 3.14.1, 3.14.2, 3.14.3, 3.14.4) presents the logical sequence of operations in the system. It starts with initialization of the ESP32, followed by sensor data reading and display updates.

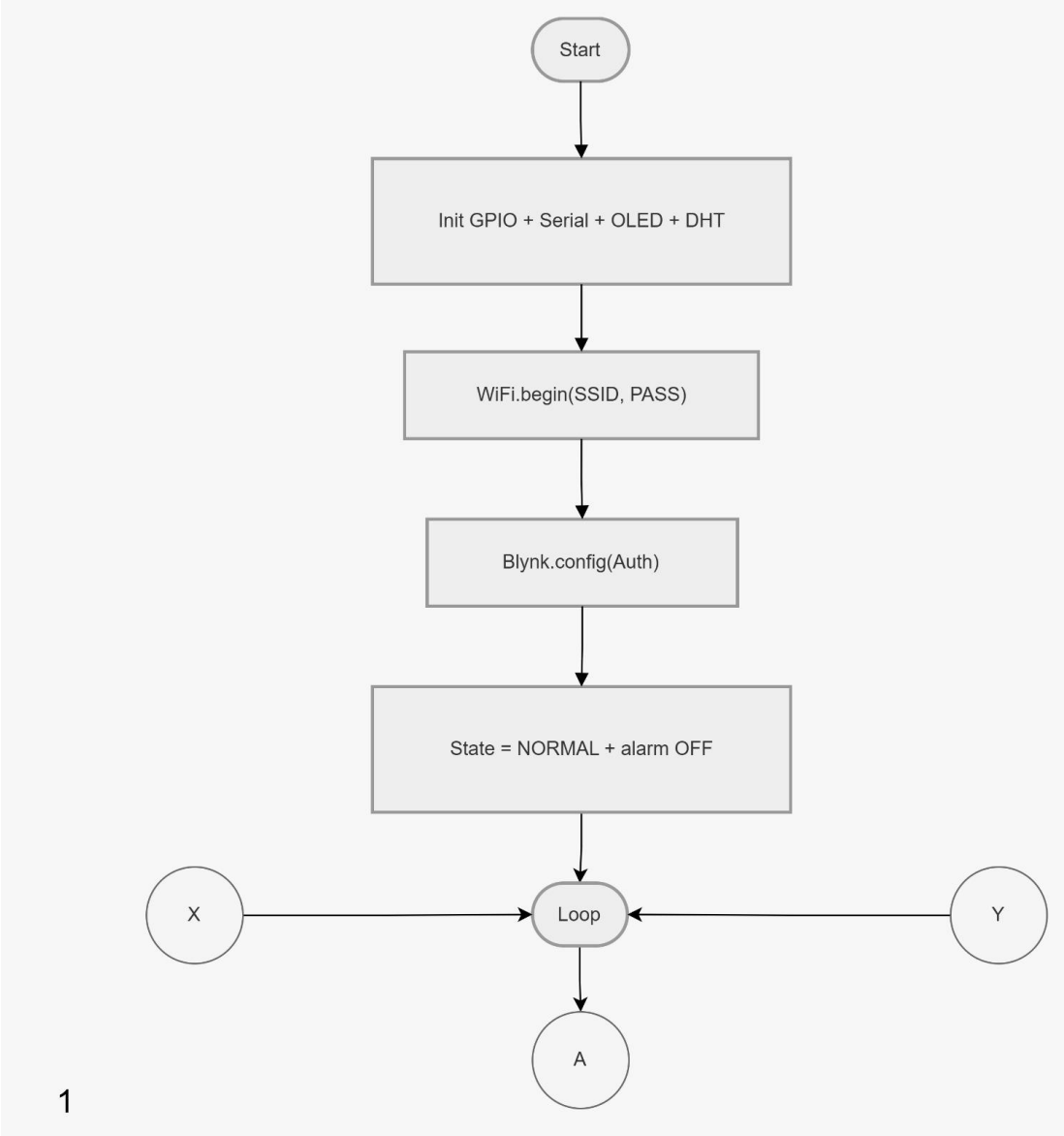


Figure 3.14.1 The Flowchart Diagram

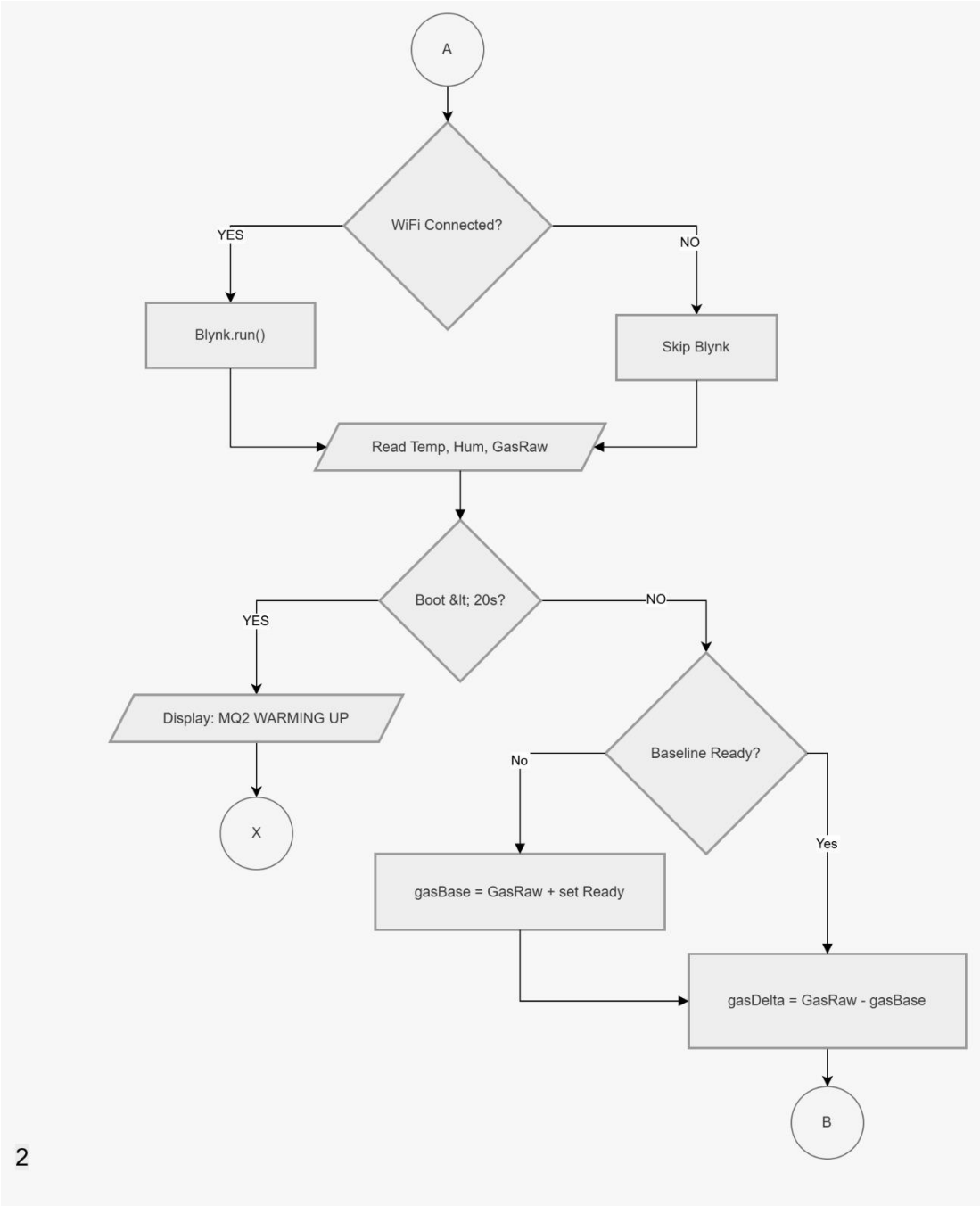


Figure 3.14.2 The Flowchart Diagram

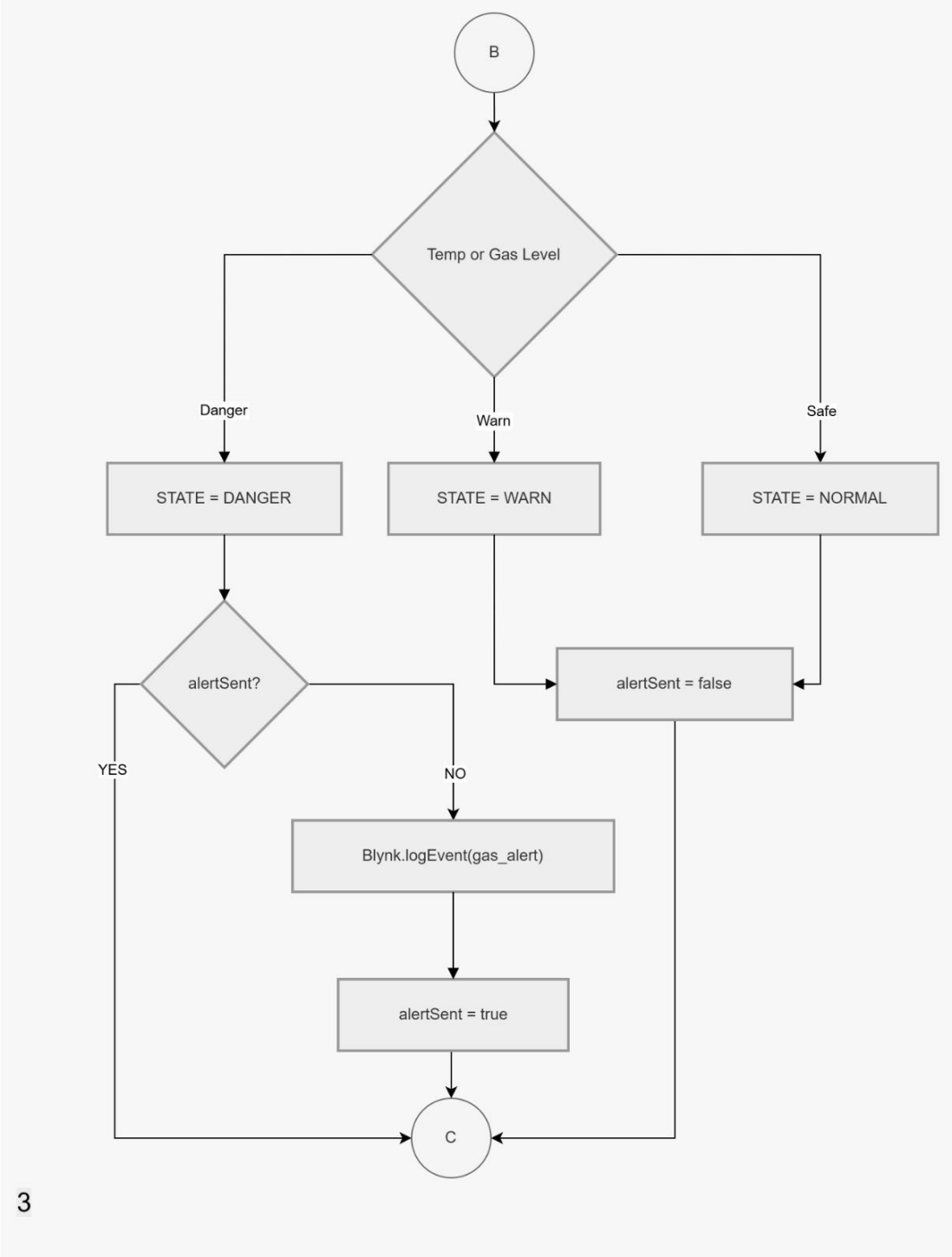


Figure 3.14.3 The Flowchart Diagram

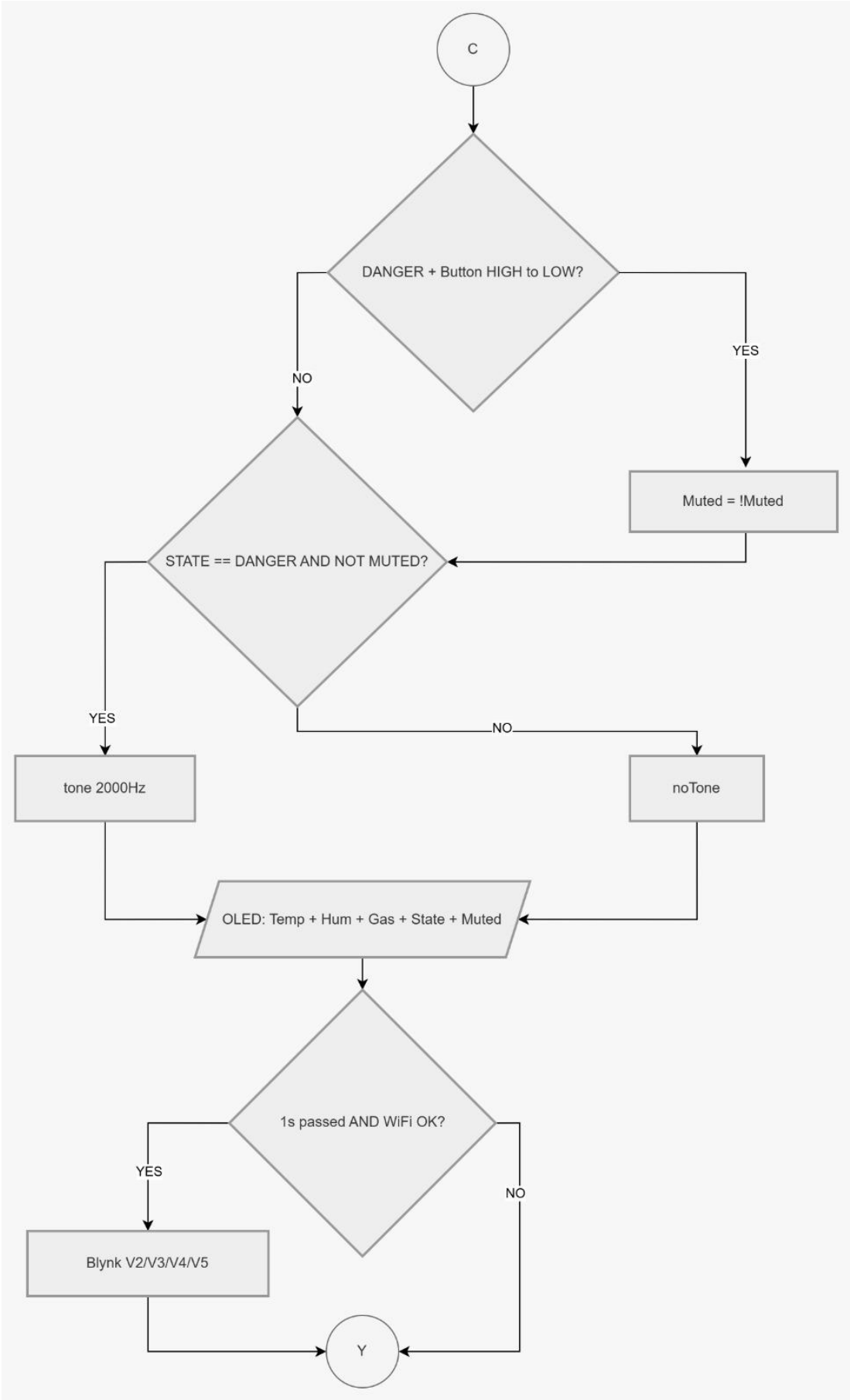


Figure 3.14.4 The Flowchart Diagram

3.9 Architecture Diagram

The system consists of input sensors, a processing unit, and output devices. As shown in Figure 3.15 (The Architecture Diagram), the MQ2 gas sensor and DHT22 temperature sensor send data to the ESP32 microcontroller, which processes the readings and determines the system state (Normal, Warning, or Danger). Based on this decision, the ESP32 controls the OLED display, LED indicators, and buzzer alarm to inform and alert the user.

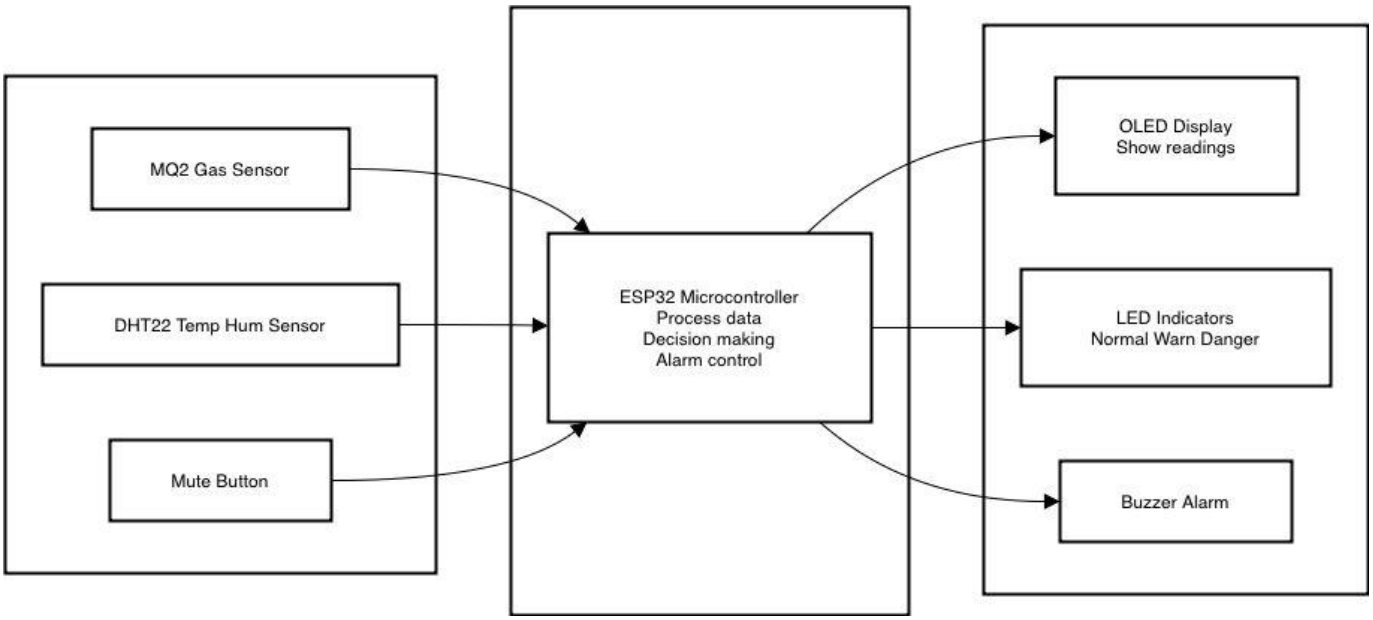


Figure 3.15 The Architecture Diagram

3.10 Software

The software of the proposed system is responsible for reading sensor data, processing the measurements, determining the system state, and controlling both local alerts and IoT notifications. The implementation was developed and tested on the ESP32 platform.

1. **Programming Language: C/C++ (Arduino framework)**
   The system logic was implemented using C/C++ within the Arduino programming framework. The code handles sensor sampling, threshold comparison, state classification (Normal/Warning/Danger), and output control (LEDs, buzzer, OLED).
2. **Development Environment (IDE): Arduino IDE**

   Arduino IDE was used to write, compile, and upload the program to the ESP32. It was selected because it provides strong ESP32 support, a large set of libraries, and fast development for embedded IoT applications.

3. **IoT/Cloud Platform: Blynk IoT**

   Blynk IoT was used to enable remote monitoring and notifications. The ESP32 connects to the Wi-Fi network and communicates with Blynk to send alert messages and (optionally) sensor readings. This allows the user to receive instant notifications when gas or temperature exceeds the defined safety thresholds.

4. **System Behavior in Software:**
   - Reads gas (MQ-2 analog) and temperature/humidity (DHT22 digital) data periodically
   - Applies warm-up and baseline calibration for the MQ-2 to reduce false alarms
   - Uses warning and danger thresholds with threshold-based decision logic
   - Updates the OLED display with real-time values and current system state
   - Activates LEDs and buzzer when abnormal conditions are detected
   - Sends IoT alerts via Blynk during dangerous conditions

3.11 Conclusion

In this chapter, the design and architecture of the proposed *Fire and Gas Detection and Alert System* were comprehensively presented. The chapter detailed the system's hardware and software integration, focusing on the functional interaction between the ESP32 microcontroller, sensors (MQ-2 and DHT22), OLED display, buzzer, and the IoT communication layer. The use of a state machine model and corresponding flowchart provided a clear representation of the system's logical behavior under various environmental conditions, ensuring reliability, responsiveness, and safety.

Furthermore, the design highlighted a modular and scalable architecture, combining real-time sensing, local alerting, and remote IoT-based notifications. Each hardware component was selected based on performance, cost-efficiency, and suitability for continuous monitoring applications. The software implementation using Arduino IDE and IoT platform supports seamless data transmission and user interaction, enhancing the overall effectiveness of the system. Overall, the design phase established a strong foundation for the practical implementation and testing of the system.

In the next chapter, we will present the results and discussion, demonstrating the system's performance, reliability, and effectiveness in detecting and alerting against fire and gas hazards.

Chapter 4
RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results and discussion of the proposed gas detection and monitoring system after implementation and testing. The evaluation aims to verify that the system can detect abnormal gas/smoke and temperature conditions, classify them correctly into operational states (Normal, Warning, and Danger), and generate appropriate responses through local indicators and remote notifications. The testing process focuses on system behavior in real-time monitoring, including sensor stability, response speed, and the clarity of user feedback through the OLED interface. In addition, the chapter discusses the effect of software features such as the MQ-2 warm-up stage. Figure 4.1 illustrates the overall testing setup and the main components used during the evaluation.

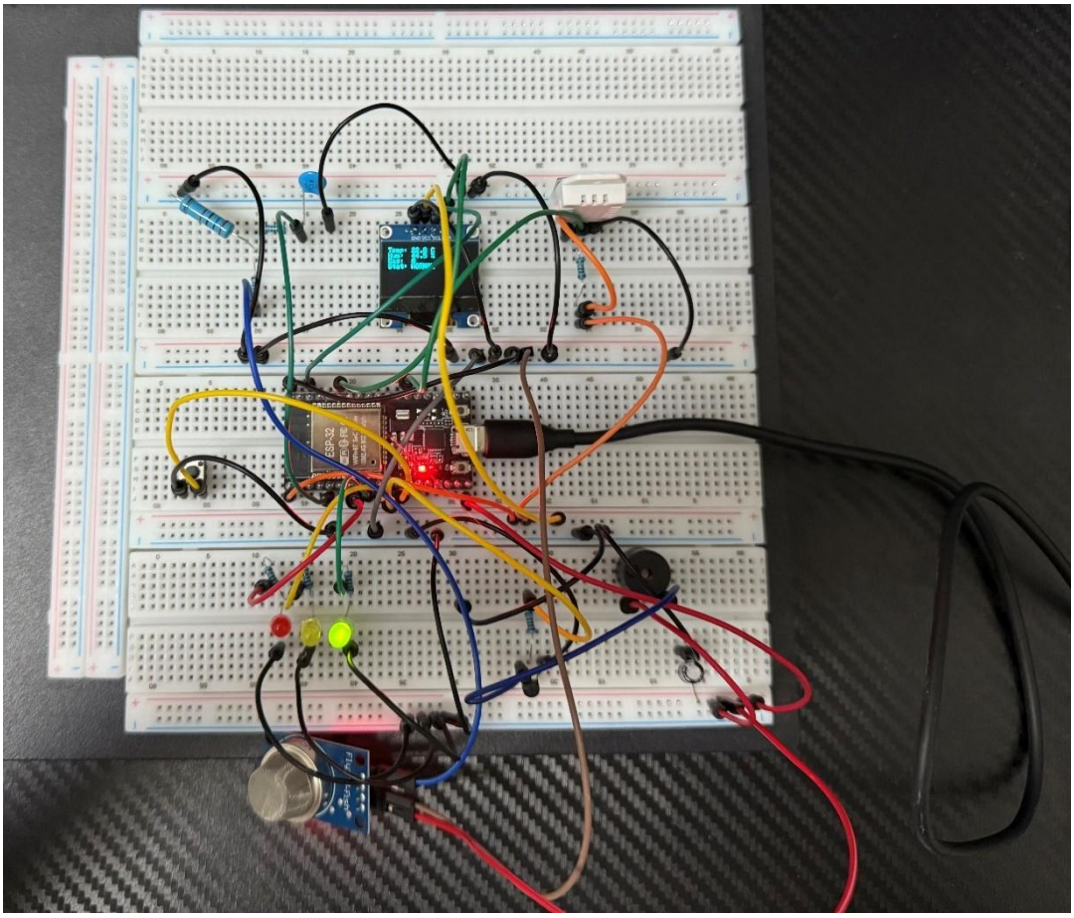


Figure 4.1 Hardware Testing Setup of the Gas Detection and Monitoring System

4.2 The Monitoring Platform in Blynk

Blynk IoT was used as the monitoring and notification platform to support remote supervision of the system. Once connected to Wi-Fi, the ESP32 can transmit status information and sensor readings to the Blynk dashboard, allowing the user to observe the system from a mobile application. The dashboard can be configured to display key values such as gas level, temperature, humidity, and the current system state, providing a clear overview of environmental conditions in real time. In addition to visualization, Blynk enables push notifications that alert the user immediately when the system detects unsafe conditions, which improves safety when the user is away from the device. This remote monitoring feature adds an important IoT layer to the project by extending system awareness beyond local alarms. Figure 4.2 shows the Blynk interface used to display readings and receive alert notifications

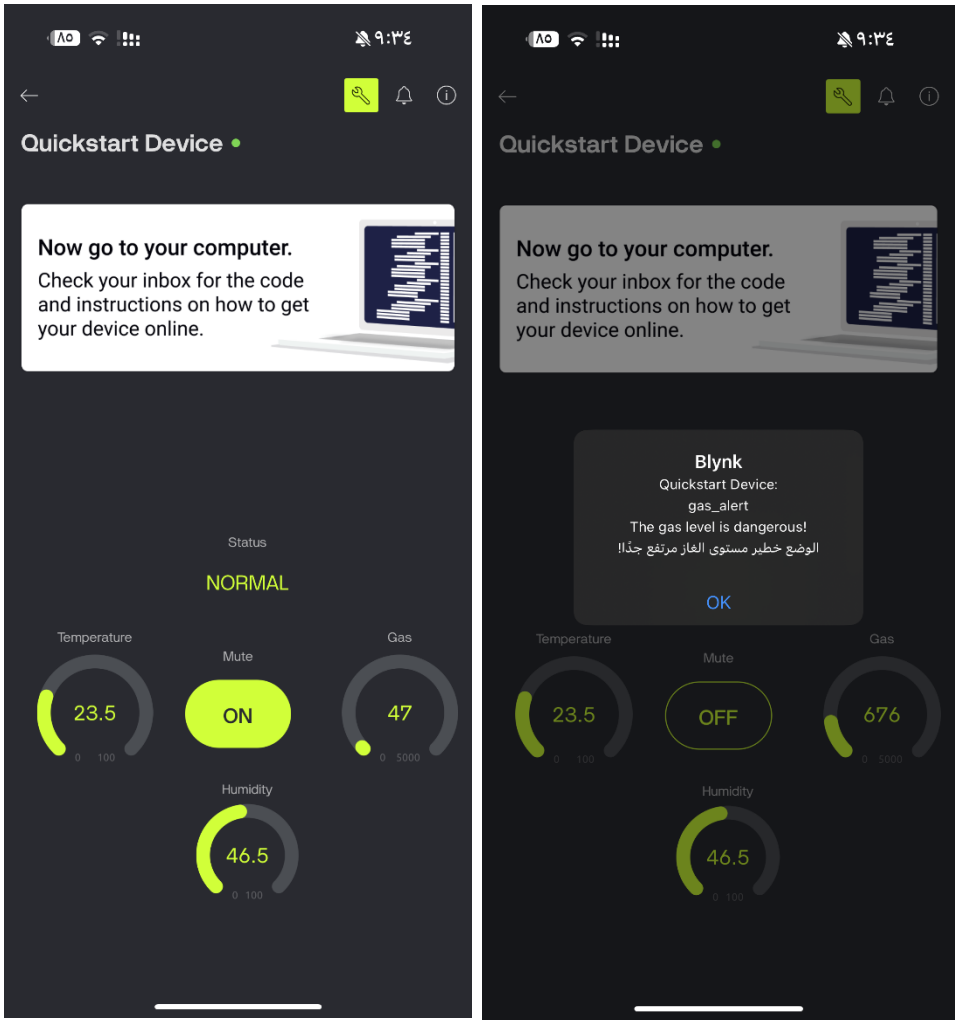


Figure 4.2 Blynk IoT Dashboard Interface for Real-Time Monitoring

4.3 Results and Discussion

The experimental observations indicate that the system responded correctly to different environmental conditions and successfully transitioned between Normal, Warning, and Danger states. Under normal conditions, sensor readings remained within safe limits, the green LED indicated a Normal state, and the buzzer stayed off while the OLED displayed stable real-time values. When the measured gas level (relative to baseline) or temperature exceeded the warning threshold, the system switched to the Warning state, activated the yellow LED, and continued monitoring while showing warning status on the OLED. When danger thresholds were exceeded, the system entered the Danger state, immediately activated the red LED and buzzer, and displayed alert messages such as gas leak or high temperature warnings, while also sending an IoT notification through Blynk. The warm-up period prevented early false triggering at startup, and the baseline calibration improved gas detection accuracy by referencing a “clean-air” value instead of relying only on raw readings. Overall, the results demonstrate that combining state-based decision logic with hysteresis produces stable behavior without frequent flickering between states. Figures 4.3.1 ,4.3.2 , and 4.3.3 present example system outputs during Normal, Warning, and Danger conditions, including OLED messages and Blynk notifications.

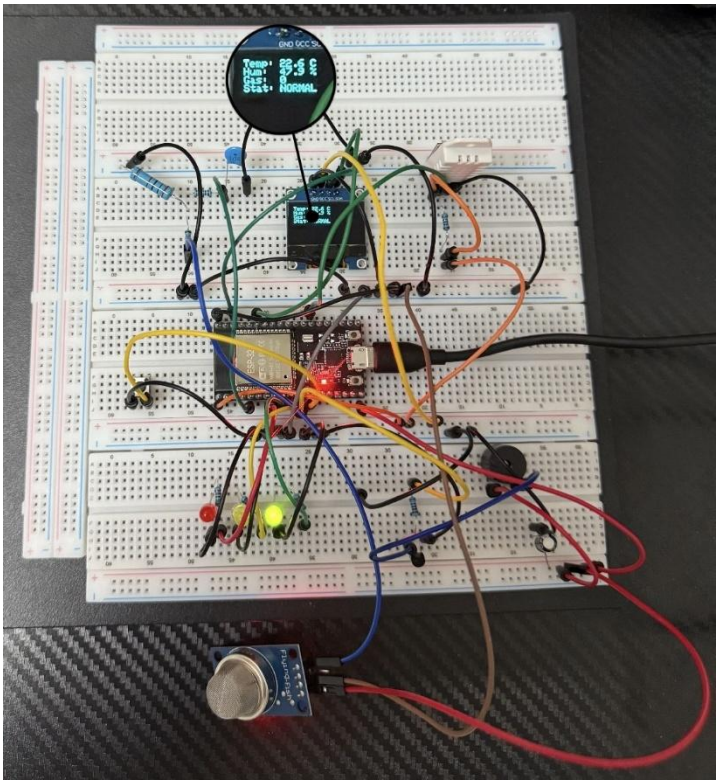


Figure 4.3.1 System Output in Normal State

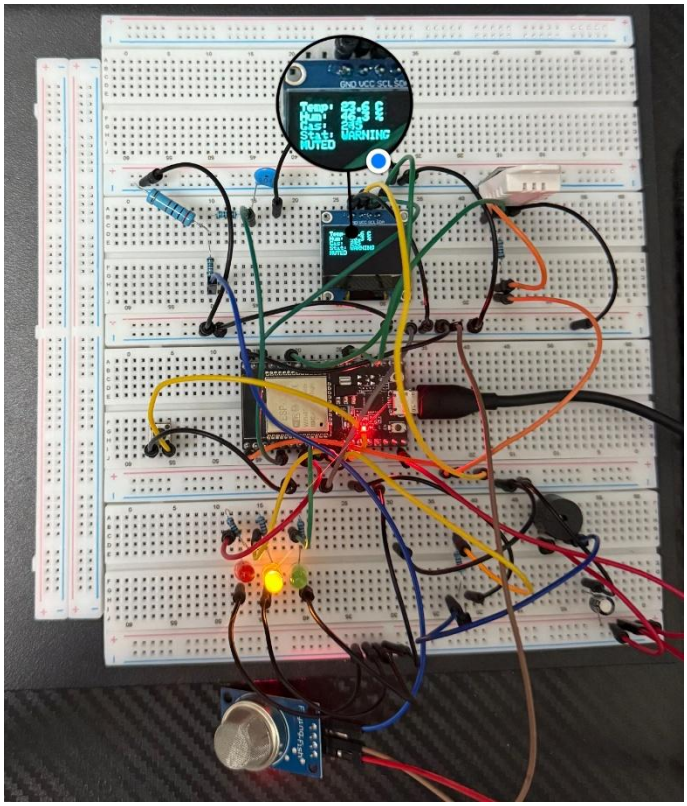


Figure 4.3.2 System Output in Warning State

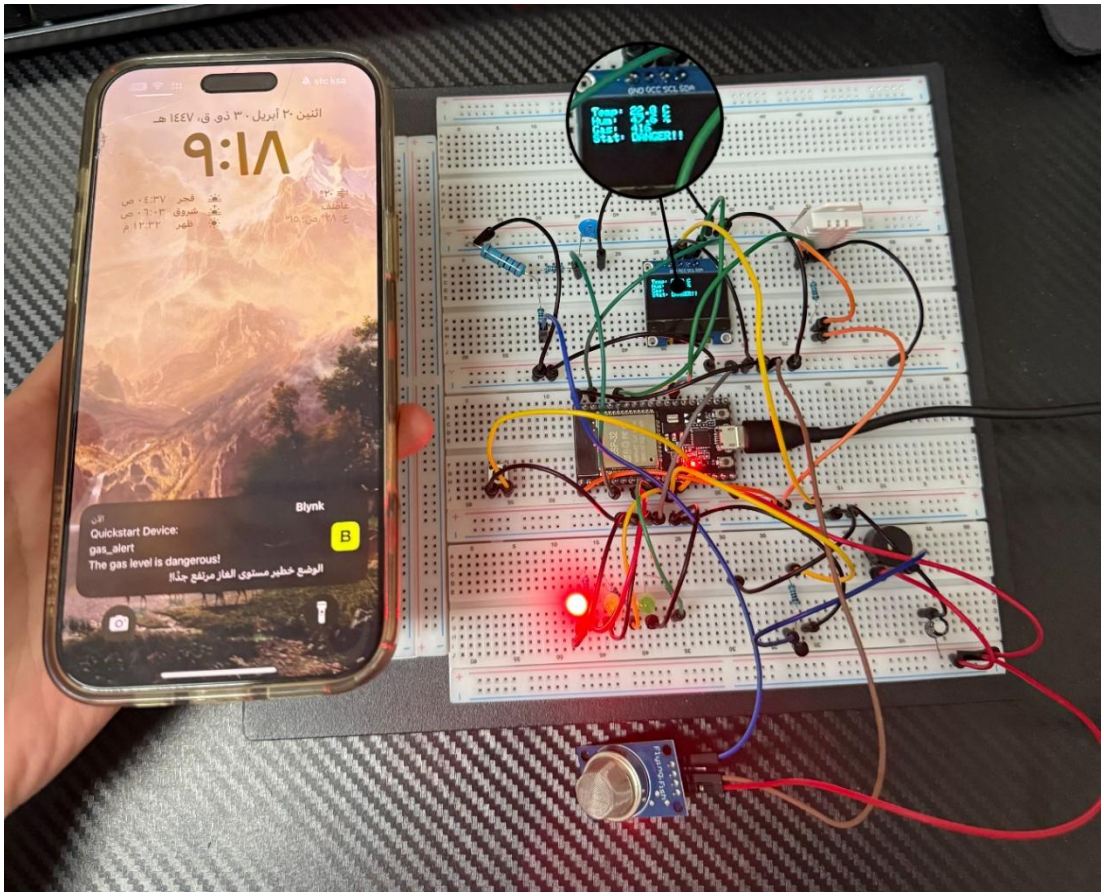


Figure 4.3.3 System Output in Danger State with IoT Notification

In Figure 4.3.1 (System Output is in Normal State) The system operates under safe environmental conditions where both gas concentration and temperature remain within predefined normal limits. In this state, the green LED is activated to indicate normal operation, while the buzzer remains off. The OLED display shows stable real-time readings of temperature, humidity, and gas level, along with a “NORMAL” status message. No alerts or notifications are generated in this condition, ensuring energy-efficient and quiet operation.

In Figure 4.3.2 (System Output is in Warning State) In this state, the system detects moderate increases in gas concentration or temperature that exceed the warning threshold but do not yet reach dangerous levels. As a result, the system transitions to the warning state, activating the yellow LED as a visual indicator. The OLED display updates to show the current readings along with a “WARNING” message, alerting the user to potential risk. However, the buzzer remains off to avoid unnecessary alarm unless conditions worsen.

In Figure 4.3.2 (System Output is in Danger State with IoT Notification) When gas concentration or temperature exceeds the defined danger threshold, the system immediately switches to the danger state. In this condition, the red LED is activated and the buzzer sounds to provide a clear and immediate local alarm. The OLED display shows a “DANGER” alert along with real-time sensor readings. In addition, a remote notification is sent to the user through the Blynk IoT application, ensuring that the user is informed even when away from the system. This integrated response enhances overall safety by combining both local and remote alert mechanisms.

Chapter 5
CONCLUSION

This chapter concludes the graduation project by summarizing the main work, outcomes, and the overall contribution of the proposed Gas Detection and Monitoring System. The system was designed and implemented using the ESP32 microcontroller as the main processing unit, along with an MQ-2 gas/smoke sensor and a DHT22 temperature and humidity sensor. The collected sensor readings were processed in real time to classify the environment into three operating states: Normal, Warning, and Danger. Based on these states, the system provided clear local alerts through LED indicators, an OLED display, and an audible buzzer alarm.

The results demonstrate that the system can effectively support early detection of hazardous gas leaks and abnormal temperature conditions. The inclusion of a warm-up stage and baseline calibration for the MQ-2 sensor helped reduce false alarms and improved the stability of gas detection decisions. In addition, the system integrated IoT functionality through ESP32 Wi-Fi and the Blynk platform, enabling remote monitoring and instant notifications. This feature enhances safety by ensuring that alerts can reach the user even when they are not physically near the device.

Overall, the proposed system provides a low-cost and practical safety solution suitable for small-scale environments such as homes and small businesses. Future improvements may include adding additional sensors (such as a dedicated smoke or CO sensor), implementing backup power to handle power outages, and integrating preventive actions such as automatic ventilation control or gas valve shutoff. These enhancements would further increase reliability and expand the system’s real-world applicability.

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